

Laboratory Work 3

Task 1

--LAB3_TASK1 Generate and insert 200 random rows

```
INSERT INTO Airport(  
airport_id,  
airport_name,  
country,  
state,  
city,  
created_at,  
updated_at  
)  
SELECT  
1000 + gs,  
'Airport' || gs,  
(ARRAY[  
'Kazakhstan',  
'United Kingdom',  
'Japan',  
'France',  
'Germany',  
'Turkey'  
])[floor(random()*6+1)::int],  
'State' || floor(random()*30+1)::int,  
'City' || gs,  
TIMESTAMP '2022-01-01'  
+ random()*(TIMESTAMP '2025-12-31' - TIMESTAMP '2022-01-01'),  
CURRENT_TIMESTAMP
```

```

FROM generate_series(1,200) AS gs;

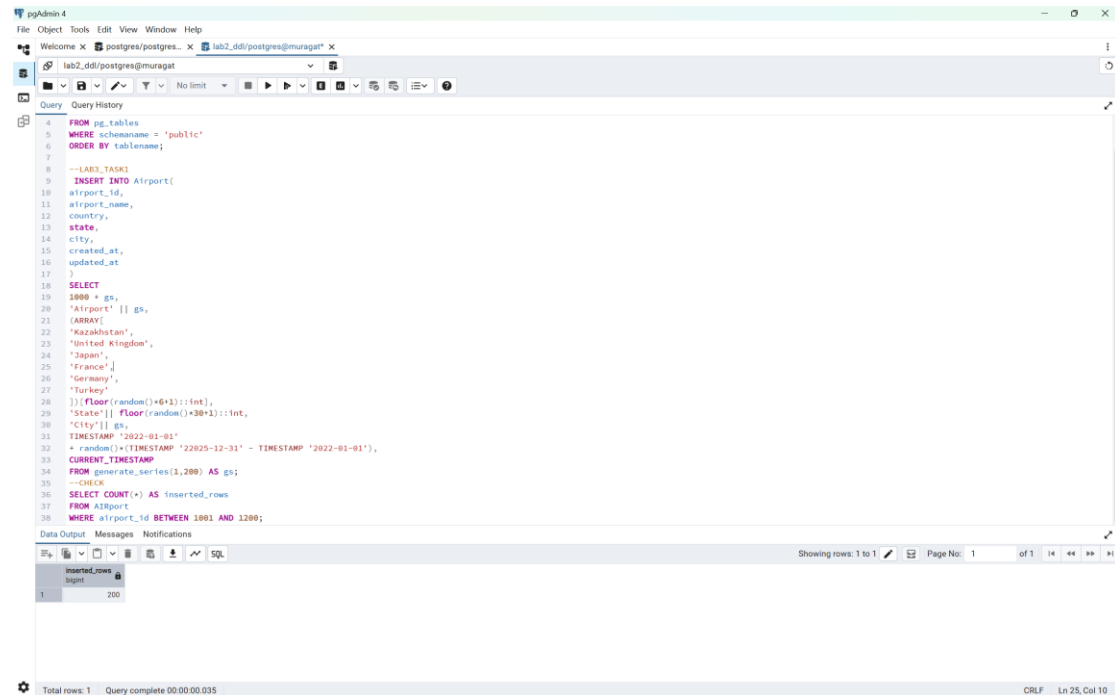
--CHECK

SELECT COUNT(*) AS inserted_rows

FROM Airport

WHERE airport_id BETWEEN 1001 AND 1200;

```



Task 2

--TASK2 Add KazAir

INSERT INTO Airline(

airline_id,

airline_code,

airline_name,

airline_country,

created_at,

updated_at

)

VALUES(

100,

'KAZ',

```

'KazAir',

'Kazakhstan',

CURRENT_TIMESTAMP,

CURRENT_TIMESTAMP

);

--CHECK

SELECT*

FROM Airline

WHERE airline_name ='KazAir';

```

The screenshot shows the pgAdmin 4 interface. The SQL query editor contains the following code:

```

30: 'Germany',
31: 'Turkey'
32: ])(floor(random()*641)::int),
33: 'State' || floor(random()*30+1)::int,
34: 'City' || gs,
35: TIMESTAMP '2022-01-01'
36: + random()*(TIMESTAMP '2025-12-31' - TIMESTAMP '2022-01-01'),
37: CURRENT_TIMESTAMP
38: FROM generate_series(1,200) AS gs;
39: --CHECK
40: SELECT COUNT(*) AS inserted_rows
41: FROM Airport
42: WHERE airport_id BETWEEN 1001 AND 1200;
43:
44: --TASK3 Add KazAir
45: INSERT INTO Airline(
46: airline_id,
47: airline_code,
48: airline_name,
49: airline_country,
50: created_at,
51: updated_at
52: )
53: VALUES(
54: 100,
55: 'KAZ',
56: 'KazAir',
57: 'Kazakhstan',
58: CURRENT_TIMESTAMP,
59: CURRENT_TIMESTAMP
60: );
61: --CHECK
62: SELECT*
63: FROM Airline
64: WHERE airline_name ='KazAir';

```

The Data Output pane shows the results of the query:

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
100	KAZ	KazAir	Kazakhstan	2026-09-29 19:48:15.103308	2026-09-29 19:48:15.103308

Total rows: 1 Query complete 00:00:00.071

Task 3

```

--TASK3 Update KazAir country to Turkey

UPDATE Airline

SET airline_country ='Turkey',

updated_at = CURRENT_TIMESTAMP

WHERE airline_name = 'KazAir';

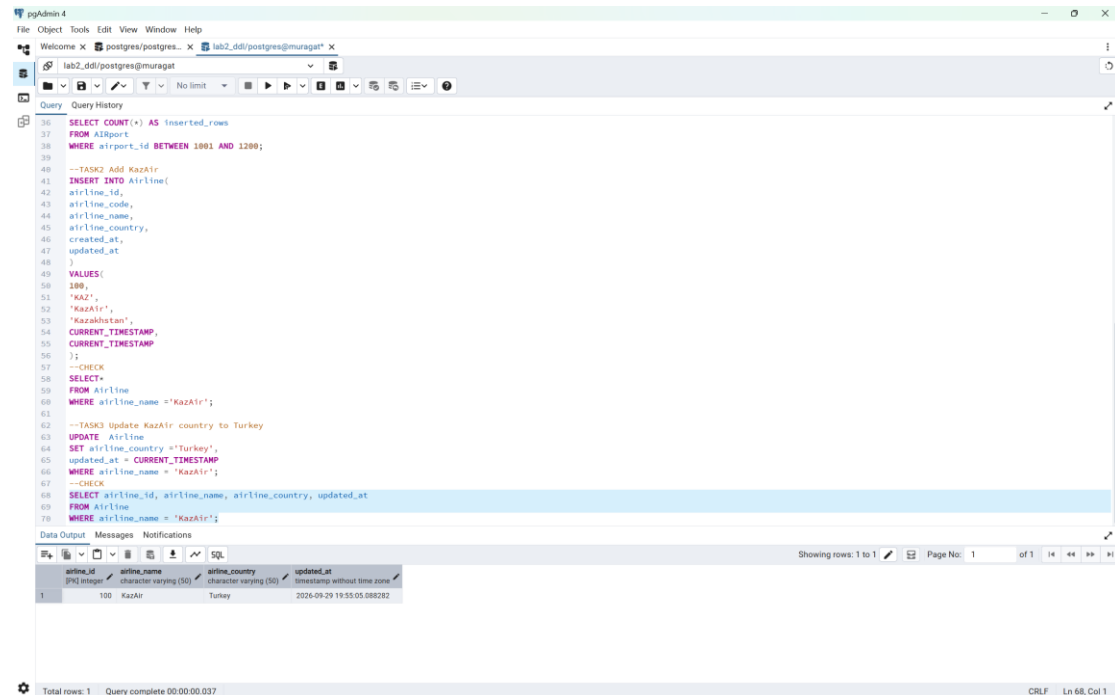
--CHECK

SELECT airline_id, airline_name, airline_country, updated_at

FROM Airline

```

WHERE airline_name = 'KazAir';



The screenshot shows the pgAdmin 4 interface. The main window displays a SQL query with the following content:

```
--TASK2 Add KazAir
INSERT INTO Airline(
  airline_id,
  airline_code,
  airline_name,
  airline_country,
  created_at,
  updated_at
)
VALUES(
  100,
  'KAZ',
  'KazAir',
  'Kazakhstan',
  CURRENT_TIMESTAMP,
  CURRENT_TIMESTAMP
);
--CHECK
SELECT
  FROM Airline
  WHERE airline_name = 'KazAir';
--TASK3 Update KazAir country to Turkey
UPDATE Airline
SET airline_country = 'Turkey',
    updated_at = CURRENT_TIMESTAMP
WHERE airline_name = 'KazAir';
--CHECK
SELECT airline_id, airline_name, airline_country, updated_at
FROM Airline
WHERE airline_name = 'KazAir';
```

The bottom panel shows the query results in a table with the following data:

airline_id	airline_name	airline_country	updated_at
100	KazAir	Turkey	2026-09-29 19:55:05.088282

The status bar at the bottom indicates "Total rows: 1" and "Query complete 00:00:00.037".

Task4

-- Task 4: Add three airlines at once

INSERT INTO Airline (

airline_id,

airline_code,

airline_name,

airline_country,

created_at,

updated_at

)

VALUES

(101, 'AE', 'AirEasy', 'France', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),

(102, 'FH', 'FlyHigh', 'Brazil', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),

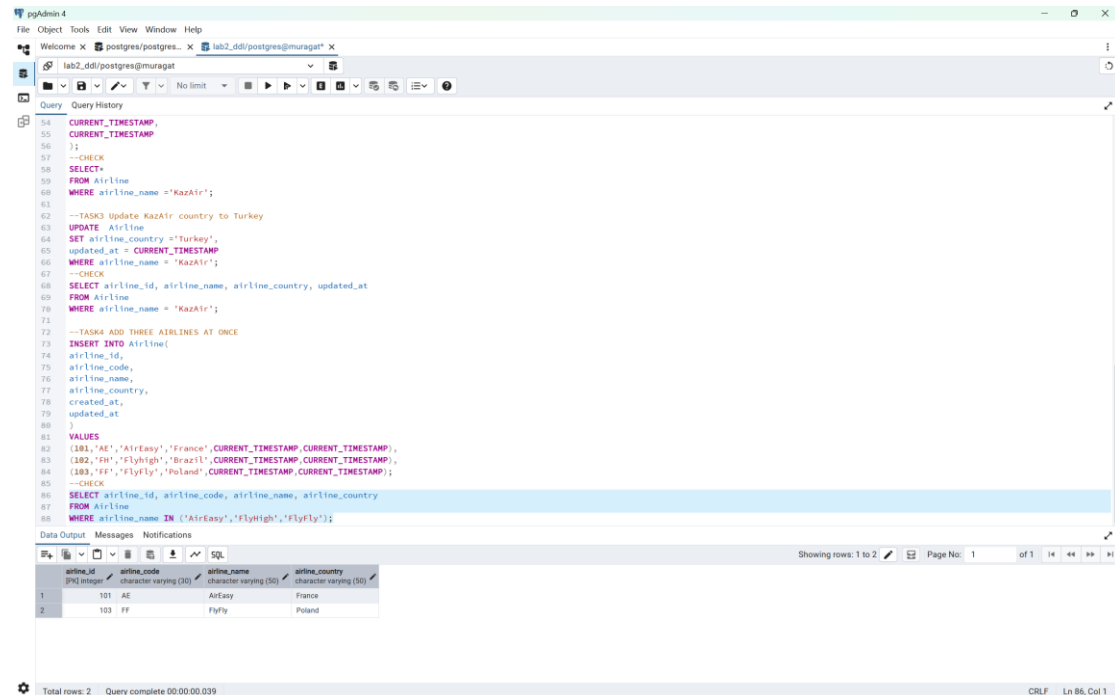
(103, 'FF', 'FlyFly', 'Poland', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);

--CHECK

SELECT airline_id, airline_code, airline_name, airline_country

FROM Airline

WHERE airline_name IN ('AirEasy', 'FlyHigh', 'FlyFly');



The screenshot shows the pgAdmin 4 interface. The main window displays a SQL query with the following content:

```
--TASK5 Update KazAir country to Turkey
UPDATE Airline
SET airline_country = 'Turkey',
updated_at = CURRENT_TIMESTAMP
WHERE airline_name = 'KazAir';
--CHECK
SELECT*
FROM Airline
WHERE airline_name = 'KazAir';
--TASK6 ADD THREE AIRLINES AT ONCE
INSERT INTO Airline(
airline_id,
airline_code,
airline_name,
airline_country,
created_at,
updated_at
)
VALUES
(101, 'AE', 'AirEasy', 'France', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
(102, 'ZH', 'FlyHigh', 'Brazil', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
(103, 'FF', 'FlyFly', 'Poland', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
--CHECK
SELECT airline_id, airline_code, airline_name, airline_country
FROM Airline
WHERE airline_name IN ('AirEasy', 'FlyHigh', 'FlyFly');
```

The Data Output tab shows the results of the query:

airline_id	airline_code	airline_name	airline_country
101	AE	AirEasy	France
103	FF	FlyFly	Poland

The status bar at the bottom indicates "Total rows: 2" and "Query complete 00:00:00.039".

Task5

--TASK5

--CHECK FLIGHTS ARRIVING IN 2024

SELECT*

FROM Flights

WHERE EXTRACT(YEAR FROM sch_arrival_time) = 2024;

-- DELETE FLIGHTS ARRIVING IN 2024

DELETE FROM Flights

WHERE EXTRACT(YEAR FROM sch_arrival_time) = 2024;

--CHECK

SELECT COUNT(*) AS remaining_2024_flights

FROM Flights

WHERE EXTRACT(YEAR FROM sch_arrival_time) = 2024;

The screenshot shows the pgAdmin 4 web interface. The top menu bar includes File, Object, Tools, Edit, View, Window, and Help. Below the menu is a toolbar with various icons for file operations and query execution. The main area is divided into two panes. The left pane shows the 'Query' tab with a SQL script. The right pane shows the 'Query History' tab. The SQL script is as follows:

```
67 --CHECK
68 SELECT airline_id, airline_name, airline_country, updated_at
69 FROM Airline
70 WHERE airline_name = 'KazAir';
71
72 -- Task 4: Add three airlines at once
73 INSERT INTO Airline (
74     airline_id,
75     airline_code,
76     airline_name,
77     airline_country,
78     created_at,
79     updated_at
80 )
81 VALUES
82 (181, 'AE', 'AirEasy', 'France', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
83 (182, 'FH', 'FlyHigh', 'Brazil', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
84 (183, 'FF', 'FlyFly', 'Poland', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
85 --CHECK
86 SELECT airline_id, airline_code, airline_name, airline_country
87 FROM Airline
88 WHERE airline_name IN ('AirEasy', 'FlyHigh', 'FlyFly');
89
90 --TASKS
91 --CHECK FLIGHTS ARRIVING IN 2024
92 SELECT*
93 FROM Flights
94 WHERE EXTRACT(YEAR FROM sch_arrival_time) = 2024;
95 -- DELETE FLIGHTS ARRIVING IN 2024
96 DELETE FROM Flights
97 WHERE EXTRACT(YEAR FROM sch_arrival_time) = 2024;
98 --CHECK
99 SELECT COUNT(*) AS remaining_2024_flights
100 FROM Flights
101 WHERE EXTRACT(YEAR FROM sch_arrival_time) = 2024;
```

The bottom pane shows the 'Data Output' tab with a table titled 'remaining_2024_flights'. The table has one row with the value 0.

remaining_2024_flights
0

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.042'.

Task6

--TASK6 INCREASE ALL TICKET PRICES BY 15%

UPDATE Booking

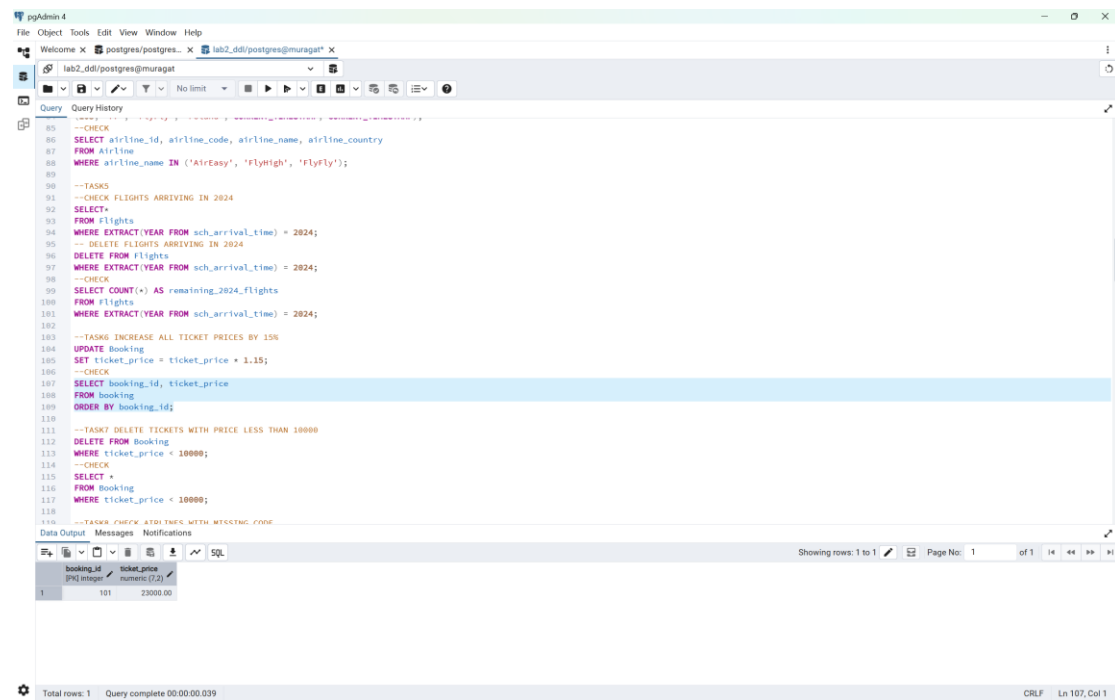
SET ticket_price = ticket_price * 1.15;

--CHECK

SELECT booking_id, ticket_price

FROM booking

ORDER BY booking_id;



Task7

--TASK7 DELETE TICKETS WITH PRICE LESS THAN 10000

DELETE FROM Booking

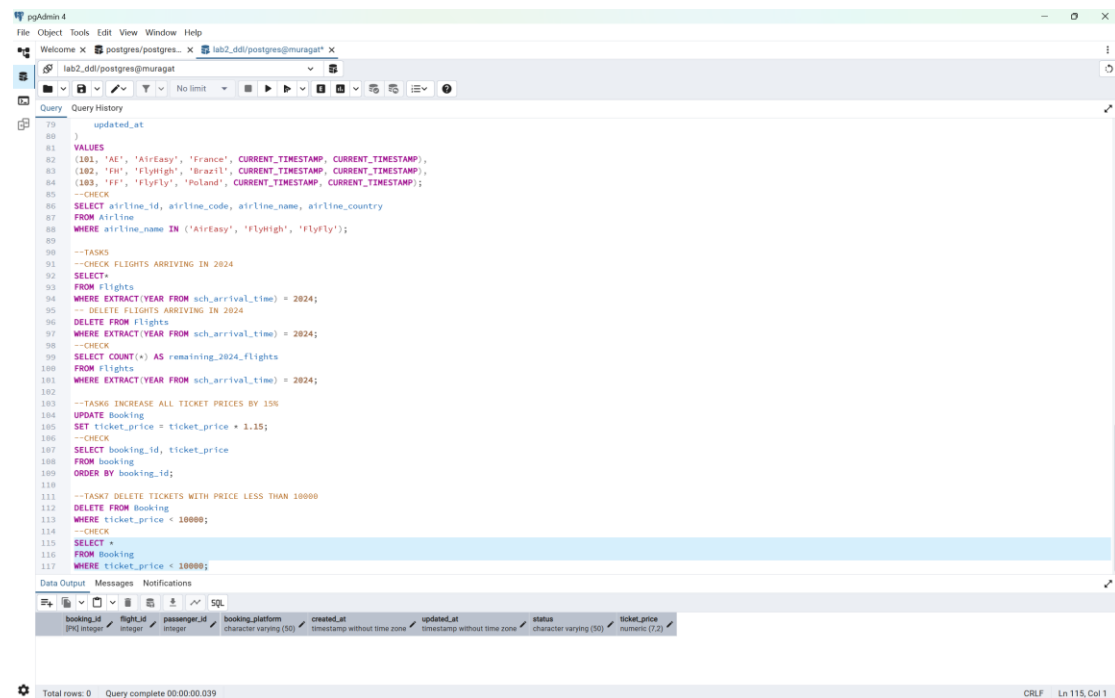
WHERE ticket_price < 10000;

--CHECK

SELECT *

FROM Booking

WHERE ticket_price < 10000;



Task8

--PREPARE TEST FOR TASK8

ALTER TABLE Airline

ALTER COLUMN airline_code DROP NOT NULL;

INSERT INTO Airline(

airline_id,

airline_code,

airline_name,

airline_country,

created_at,

updated_at

)

VALUES(

999,

NULL,

'NoCode Air',

'Kazakhstan',


```

CURRENT_TIMESTAMP,
CURRENT_TIMESTAMP
);

--TASK8 CHECK AIRLINES WITH MISSING CODE

SELECT airline_id, airline_name, airline_code

FROM Airline

WHERE airline_code IS NULL;

--SET MISSING AIRLINE CODES TO UNK

UPDATE Airline

SET airline_code = 'UNK'

WHERE airline_code IS NULL;

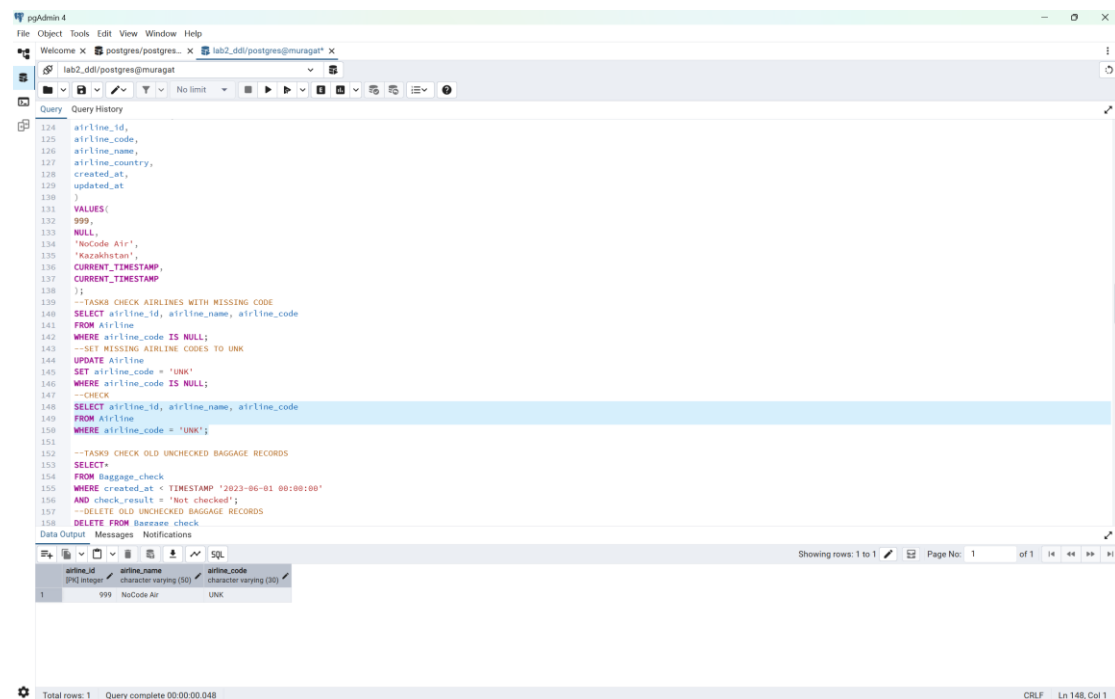
--CHECK

SELECT airline_id, airline_name, airline_code

FROM Airline

WHERE airline_code = 'UNK';

```



Task9

```

--TASK9 CHECK OLD UNCHECKED BAGGAGE RECORDS

SELECT*

```

FROM Baggage_check

WHERE created_at < TIMESTAMP '2023-06-01 00:00:00'

AND check_result = 'Not checked';

--DELETE OLD UNCHECKED BAGGAGE RECORDS

DELETE FROM Baggage_check

WHERE created_at < TIMESTAMP '2023-06-01 00:00:00'

AND check_result = 'Not checked';

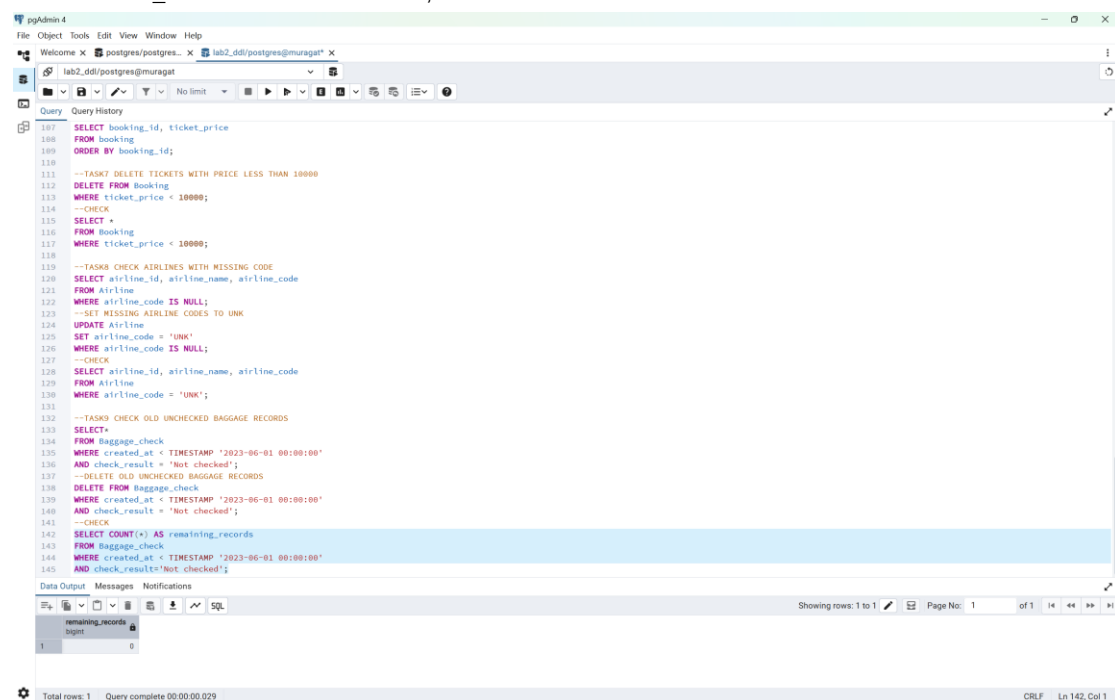
--CHECK

SELECT COUNT(*) AS remaining_records

FROM Baggage_check

WHERE created_at < TIMESTAMP '2023-06-01 00:00:00'

AND check_result='Not checked';



The screenshot shows the pgAdmin 4 web interface. The top menu bar includes File, Object, Tools, Edit, View, Window, and Help. Below the menu is a toolbar with various icons. The main area displays a SQL query with line numbers 107 to 145. The query includes several tasks: deleting tickets with price less than 10000, checking airlines with missing codes, and deleting old unchecked baggage records. The final result is shown in the 'Data Output' tab, which displays a single row with the column 'remaining_records' and the value '0'. The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.029'.

```
107 SELECT booking_id, ticket_price
108 FROM booking
109 ORDER BY booking_id;
110
111 --TASK7 DELETE TICKETS WITH PRICE LESS THAN 10000
112 DELETE FROM booking
113 WHERE ticket_price < 10000;
114
115 --CHECK
116 SELECT *
117 FROM booking
118 WHERE ticket_price < 10000;
119
120 --TASK8 CHECK AIRLINES WITH MISSING CODE
121 SELECT airline_id, airline_name, airline_code
122 FROM Airline
123 WHERE airline_code IS NULL;
124
125 --SET MISSING AIRLINE CODES TO UNK
126 UPDATE Airline
127 SET airline_code = 'UNK'
128 WHERE airline_code IS NULL;
129
130 --CHECK
131 SELECT airline_id, airline_name, airline_code
132 FROM Airline
133 WHERE airline_code = 'UNK';
134
135 --TASK9 CHECK OLD UNCHECKED BAGGAGE RECORDS
136 SELECT *
137 FROM Baggage_check
138 WHERE created_at < TIMESTAMP '2023-06-01 00:00:00'
139 AND check_result = 'Not checked';
140
141 --DELETE OLD UNCHECKED BAGGAGE RECORDS
142 DELETE FROM Baggage_check
143 WHERE created_at < TIMESTAMP '2023-06-01 00:00:00'
144 AND check_result = 'Not checked';
145
146 --CHECK
147 SELECT COUNT(*) AS remaining_records
148 FROM Baggage_check
149 WHERE created_at < TIMESTAMP '2023-06-01 00:00:00'
150 AND check_result='Not checked';
```

remaining_records
0

Task10

--TASK10 CHECK MATCHING AIRPORTS

SELECT*

FROM Airport

WHERE state IS NULL

AND city IN('Mlawe','Kepuh');

--DELETE MATCHING AIRPORTS

DELETE FROM Airport

WHERE state IS NULL

AND city IN('Mlawe','Kepuh');

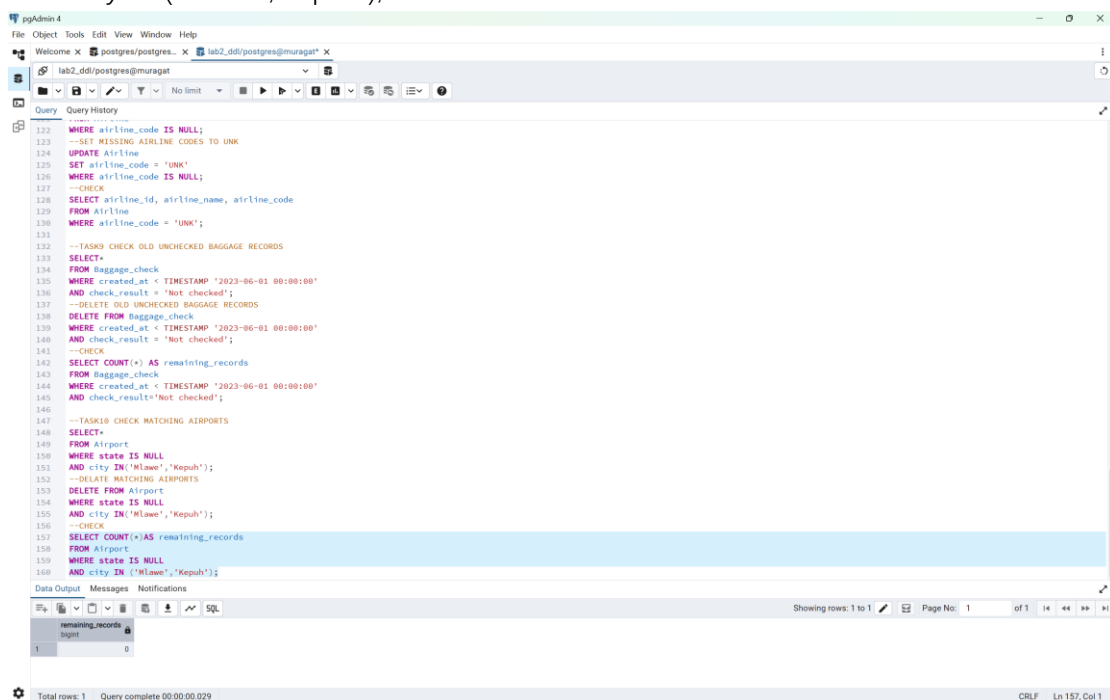
--CHECK

SELECT COUNT(*)AS remaining_records

FROM Airport

WHERE state IS NULL

AND city IN ('Mlawe','Kepuh');



The screenshot shows the pgAdmin 4 interface with a SQL query editor and a results pane. The query is as follows:

```
122 WHERE airline_code IS NULL;
123 --SET MISSING AIRLINE CODES TO UNK
124 UPDATE Airline
125 SET airline_code = 'UNK'
126 WHERE airline_code IS NULL;
127 --CHECK
128 SELECT airline_id, airline_name, airline_code
129 FROM Airline
130 WHERE airline_code = 'UNK';
131
132 --TASK9 CHECK OLD UNCHECKED BAGGAGE RECORDS
133 SELECT*
134 FROM Baggage_check
135 WHERE created_at < TIMESTAMPT '2023-06-01 00:00:00'
136 AND check_result = 'Not checked';
137 --DELETE OLD UNCHECKED BAGGAGE RECORDS
138 DELETE FROM Baggage_check
139 WHERE created_at < TIMESTAMPT '2023-06-01 00:00:00'
140 AND check_result = 'Not checked';
141 --CHECK
142 SELECT COUNT(*) AS remaining_records
143 FROM Baggage_check
144 WHERE created_at < TIMESTAMPT '2023-06-01 00:00:00'
145 AND check_result='Not checked';
146
147 --TASK10 CHECK MATCHING AIRPORTS
148 SELECT*
149 FROM Airport
150 WHERE state IS NULL
151 AND city IN('Mlawe','Kepuh');
152 --DELETE MATCHING AIRPORTS
153 DELETE FROM Airport
154 WHERE state IS NULL
155 AND city IN('Mlawe','Kepuh');
156 --CHECK
157 SELECT COUNT(*)AS remaining_records
158 FROM Airport
159 WHERE state IS NULL
160 AND city IN ('Mlawe','Kepuh');
```

The results pane shows a single row with the column 'remaining_records' and the value '0'.

remaining_records
0

Task11

--TASK11 INSERT BAGGAGE CHECK AND RETURN GENERATED VALUES

INSERT INTO Baggage_check(

baggage_check_id,

check_result,

created_at,

updated_at,

booking_id,

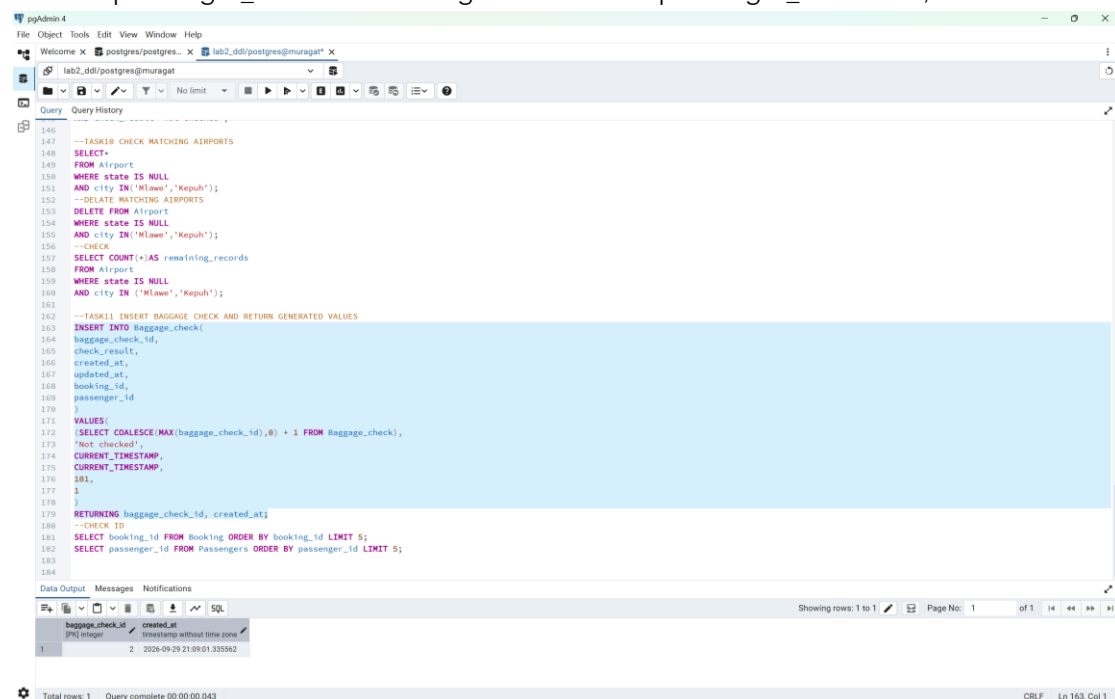
passenger_id

```

)
VALUES(
(SELECT COALESCE(MAX(baggage_check_id),0) + 1 FROM Baggage_check),
'Not checked',
CURRENT_TIMESTAMP,
CURRENT_TIMESTAMP,
101,
1
)
RETURNING baggage_check_id, created_at;

--CHECK ID
SELECT booking_id FROM Booking ORDER BY booking_id LIMIT 5;
SELECT passenger_id FROM Passengers ORDER BY passenger_id LIMIT 5;

```



The screenshot shows the pgAdmin 4 interface. The main window displays a SQL query with line numbers 146 to 184. The query includes comments for Task 10 and Task 11, and SQL code for inserting baggage check records. The results pane at the bottom shows a single row of data.

```

146 --TASK10 CHECK MATCHING AIRPORTS
147
148 SELECT
149 FROM Airport
150 WHERE state IS NULL
151 AND city IN ('Mlawe','Kepuh');
152 --DELETE MATCHING AIRPORTS
153 DELETE FROM Airport
154 WHERE state IS NULL
155 AND city IN ('Mlawe','Kepuh');
156 --CHECK
157 SELECT COUNT(*) AS remaining_records
158 FROM Airport
159 WHERE state IS NULL
160 AND city IN ('Mlawe','Kepuh');
161
162 --TASK11 INSERT BAGGAGE CHECK AND RETURN GENERATED VALUES
163 INSERT INTO baggage_check(
164 baggage_check_id,
165 check_result,
166 created_at,
167 updated_at,
168 booking_id,
169 passenger_id
170 )
171 VALUES(
172 (SELECT COALESCE(MAX(baggage_check_id),0) + 1 FROM Baggage_check),
173 'Not checked',
174 CURRENT_TIMESTAMP,
175 CURRENT_TIMESTAMP,
176 101,
177 1
178 )
179 RETURNING baggage_check_id, created_at;
180 --CHECK ID
181 SELECT booking_id FROM Booking ORDER BY booking_id LIMIT 5;
182 SELECT passenger_id FROM Passengers ORDER BY passenger_id LIMIT 5;
183
184

```

Data Output:

baggage_check_id	created_at
101	2026-09-29 21:09:01.335562

Total rows: 1 Query complete 00:00:00.043

Task12

```

--TASK12 CONVERT ALL COUNTRIES TO UPPERCASE

UPDATE Airline

SET airline_country = UPPER(airline_country);

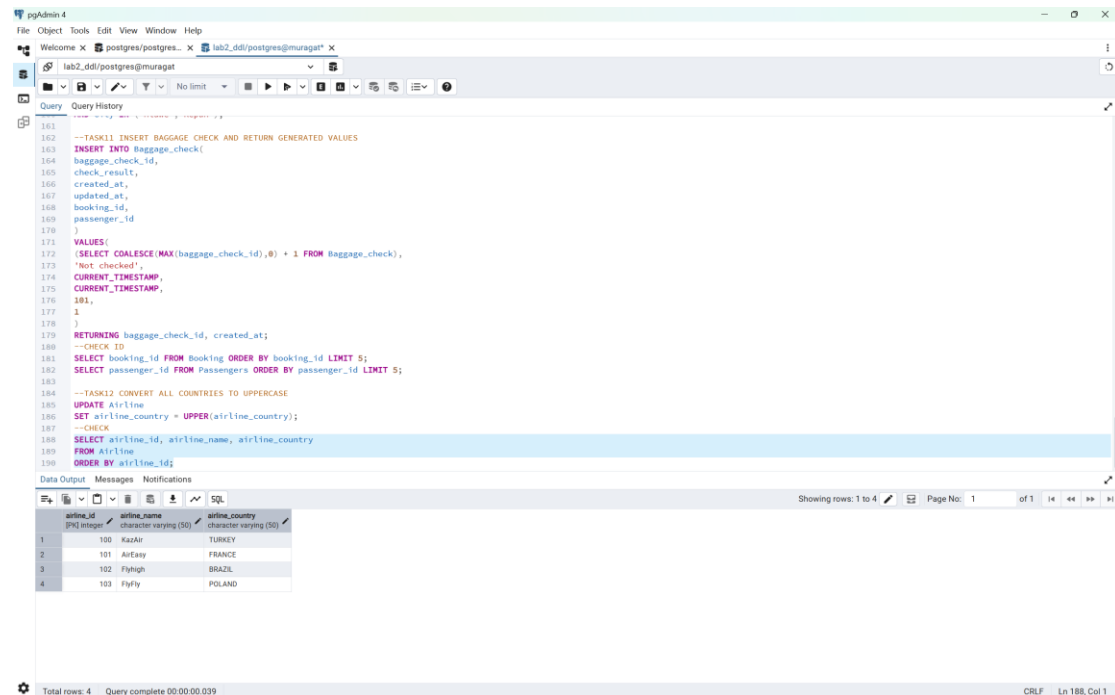
--CHECK

```

SELECT airline_id, airline_name, airline_country

FROM Airline

ORDER BY airline_id;



Task13

-- Prepare test data for Task 13

INSERT INTO Airline (

airline_id,

airline_code,

airline_name,

airline_country,

created_at,

updated_at

)

VALUES (

5,

'GA',

'Old Airline',

'Old Country',

```

CURRENT_TIMESTAMP,
CURRENT_TIMESTAMP
);

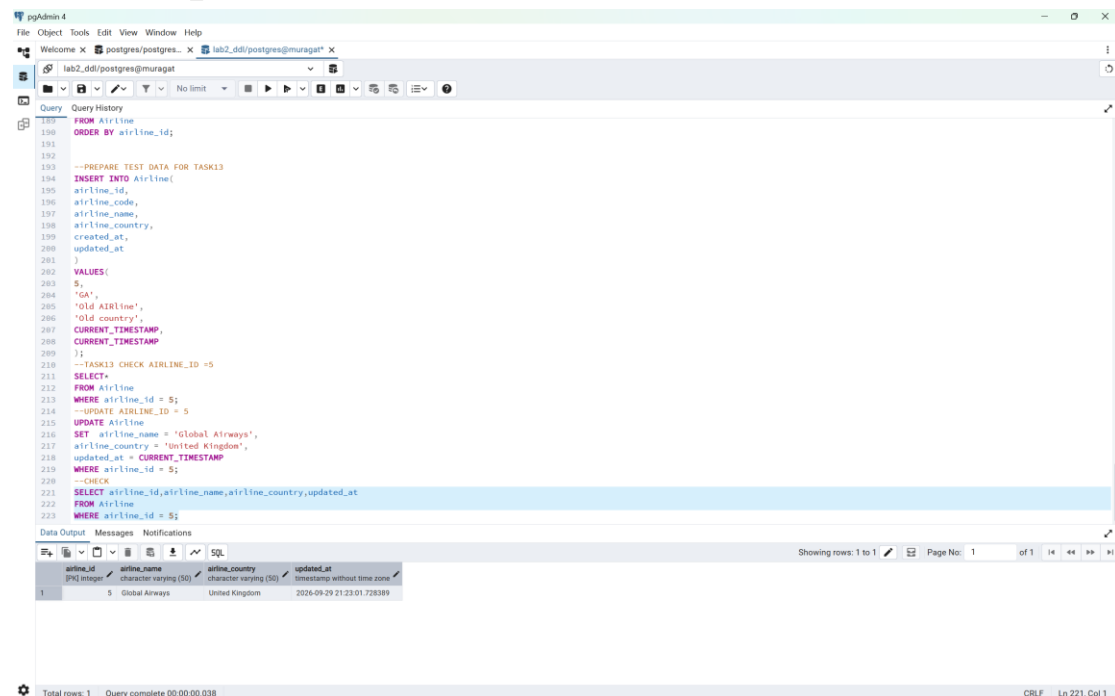
-- Task 13: Check airline_id = 5

SELECT *
FROM Airline
WHERE airline_id = 5;

-- CHECK TASK 13

SELECT airline_id, airline_name, airline_country, updated_at
FROM Airline
WHERE airline_id = 5;

```



Task14

```
--PREPARE TEST DATA FOR TASK 14
```

```

INSERT INTO Airport(
airport_id,
airport_name,
country,
state,

```

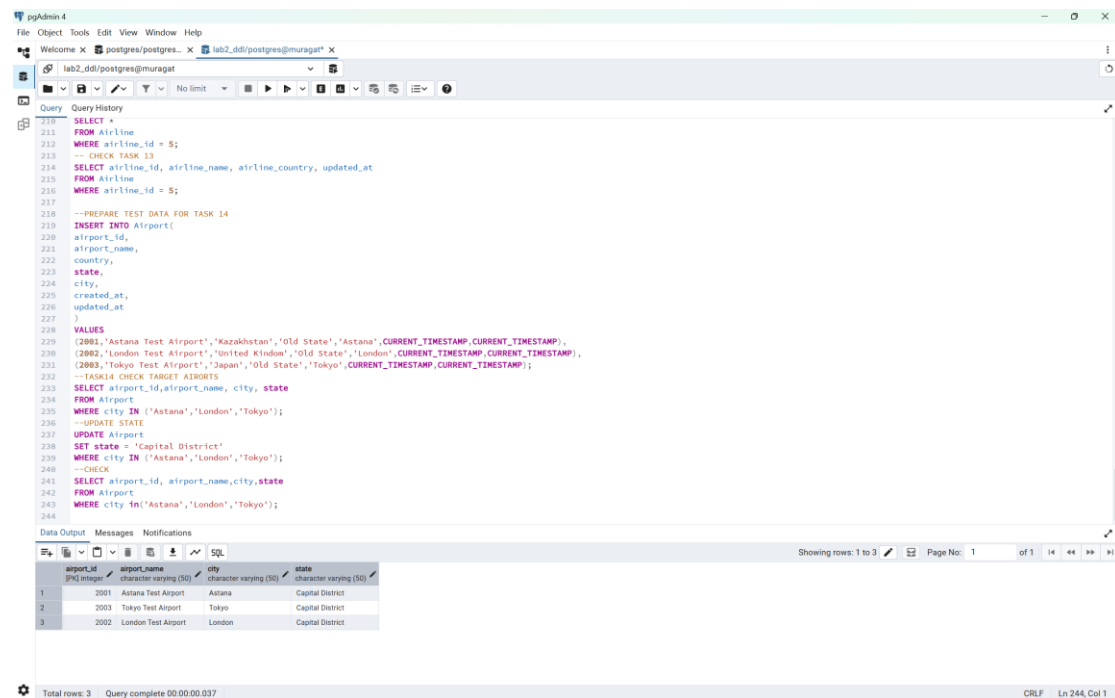
```
city,
created_at,
updated_at
)
VALUES
(2001,'Astana Test Airport','Kazakhstan','Old
State','Astana',CURRENT_TIMESTAMP,CURRENT_TIMESTAMP),
(2002,'London Test Airport','United Kindom','Old
State','London',CURRENT_TIMESTAMP,CURRENT_TIMESTAMP),
(2003,'Tokyo Test Airport','Japan','Old
State','Tokyo',CURRENT_TIMESTAMP,CURRENT_TIMESTAMP);

--TASK14 CHECK TARGET AIRORTS

SELECT airport_id,airport_name, city, state
FROM Airport
WHERE city IN ('Astana','London','Tokyo');

--UPDATE STATE
UPDATE Airport
SET state = 'Capital District'
WHERE city IN ('Astana','London','Tokyo');

--CHECK
SELECT airport_id, airport_name,city,state
FROM Airport
WHERE city in('Astana','London','Tokyo');
```



Task 15

--PREPARE TEST DATA FOR TASK 15

INSERT INTO Baggage_check(

baggage_check_id,

check_result,

created_at,

updated_at,

booking_id,

passenger_id

)

VALUES(

(SELECT COALESCE(MAX(baggage_check_id),0)+1 FROM Baggage_check),

'Not checked',

TIMESTAMP '2024-03-15 10:00:00',

TIMESTAMP '2024-03-15 10:00:00',

101,

1


```

)

RETURNING baggage_check_id,check_result,created_at;

--TASK15 CHECK MARCH 2024 UNCHECKED BAGGAGE RECORDS

SELECT*

FROM Baggage_check

WHERE created_at>=TIMESTAMP '2024-03-01 00:00:00'

AND created_at< TIMESTAMP '2024-04-01 00:00:00'

AND check_result = 'Not checked';

--MARK MARCH 2024 BAGGAGE CHECKS AS CHECKED

UPDATE Baggage_check

SET check_result='Checked',

updated_at=CURRENT_TIMESTAMP

WHERE created_at>=TIMESTAMP '2024-03-01 00:00:00'

AND created_at<TIMESTAMP '2024-04-01 00:00:00'

AND check_result= 'Not checked';

--CHECK

SELECT baggage_check_id,check_result,created_at,updated_at

FROM Baggage_check

WHERE created_at>=TIMESTAMP'2024-03-01 00:00:00'

AND created_at=TIMESTAMP '2024-04-01 00:00:00';


UPDATE Baggage_check

SET check_result ='Checked',

updated_at= CURRENT_TIMESTAMP

WHERE baggage_check_id=5;


SELECT baggage_check_id,check_result,created_at,updated_at

FROM Baggage_check

WHERE baggage_check_id=5;

```

pgAdmin 4

File Object Tools Edit View Window Help

Welcome X postgres/postgres.. X lab2_ddl/postgres@muragatt X

lab2_ddl/postgres@muragatt

No limit

Query

```
Query History
2755 (SELECT COALESCE(MAX(baggage_check_id),0)+1 FROM Baggage_check),
2756 'Not checked',
2757 TIMESTAMP '2024-03-15 10:00:00',
2758 TIMESTAMP '2024-03-15 10:00:00',
2759 101,
2760 1
2761 )
2762 RETURNING baggage_check_id,check_result,created_at;
2763 --TASK15 CHECK MARCH 2024 UNCHECKED BAGGAGE RECORDS
2764 SELECT*
2765 FROM Baggage_check
2766 WHERE created_at>=TIMESTAMP '2024-03-01 00:00:00'
2767 AND created_at<=TIMESTAMP '2024-04-01 00:00:00'
2768 AND check_result = 'Not checked';
2769 --ROSK MARCH 2024 BAGGAGE CHECKS AS CHECKED
2770 UPDATE Baggage_check
2771 SET check_result='checked',
2772 updated_at=CURRENT_TIMESTAMP
2773 WHERE created_at>=TIMESTAMP '2024-03-01 00:00:00'
2774 AND created_at<=TIMESTAMP '2024-04-01 00:00:00'
2775 AND check_result= 'Not checked';
2776 --CHECK
2777 SELECT baggage_check_id,check_result,created_at,updated_at
2778 FROM Baggage_check
2779 WHERE created_at>=TIMESTAMP '2024-03-01 00:00:00'
2780 AND created_at<=TIMESTAMP '2024-04-01 00:00:00';
2781
2782 UPDATE Baggage_check
2783 SET check_result = 'checked',
2784 updated_at= CURRENT_TIMESTAMP
2785 WHERE baggage_check_id=5;
2786
2787 SELECT baggage_check_id,check_result,created_at,updated_at
2788 FROM Baggage_check
2789 WHERE baggage_check_id=5;
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

baggage_check_id	check_result	created_at	updated_at
[PK] integer	character varying (50)	timestamp without time zone	timestamp without time zone
1	5 Checked	2024-03-15 10:00:00	2026-09-29 22:08:12.263959

Total rows: 1 Query complete 00:00:00.043 CRLF Ln 244, Col 1